

Recording a route without satellites

How accurate a phone’s own sensors are, measured against GPS

Yu Yao-Hsing
euler.yu@gmail.com

Abstract

When GPS is lost — in a tunnel, an underground car park, or an aircraft — a phone can still draw the route it travels. It reads *speed* from how the vehicle shakes and *direction* from its gyroscope and compass, and it never uses GPS after the first point. This paper measures how well the newest version of that system works, by running it over every earlier recording and comparing the result with GPS recorded at the same time, used only as the answer key. Over 70 journeys (18 hours), the total distance came out 1.5% short, and 64% of journeys were within 20%. Over 64 whole routes, the median route was turned 13° from its true direction, and 78% were within 30°. On a flight, the direction stayed within 30° of the GPS track 99% of the time. The main weaknesses are speed at the extremes — fast riding reads slow and slow riding reads fast — and short journeys, where there are too few turns to learn how the phone sits in the pocket.

1 What the system does

A route is a series of short steps. Each step needs a length and a direction. With GPS both come from satellites; here both come from the phone’s motion sensors.

Speed from vibration. A moving vehicle shakes, and it shakes differently at different speeds [1]. (Adding up acceleration instead does not work on a phone: its small sensor errors grow into kilometres within minutes [2].) Every second, the phone takes the last four seconds of up-and-down acceleration, measured 50 times a second, and turns it into a short “fingerprint” of eleven numbers — how much energy the shaking has at each of nine frequency ranges, plus two measures of its overall strength. Whenever GPS is available, the phone stores fingerprints together with the speed GPS measured at that moment; it keeps up to 4000 of them. Without GPS, it looks up the 12 stored fingerprints closest to the current one and averages their speeds:

$$\hat{v} = \frac{\sum_{j=1}^{12} v_j / d_j^2}{\sum_{j=1}^{12} 1 / d_j^2} \quad (1)$$

where v_j is a stored speed and d_j how far its fingerprint is from the current one. If even the closest stored fingerprint is far away, the phone does not guess. Averaging tends to pull every answer toward the middle, so a correction learned from the stored examples stretches the answers back out.

Direction from turns. The gyroscope measures every turn precisely but does not know north; the compass knows north but measures where the *phone* points, not where the vehicle is going — a known problem for phones carried in different ways [3]. In a pocket those differ by some angle β , which depends on how the phone happens to sit. The phone learns β from the vehicle’s own turns. In a turn the vehicle is pushed toward the inside of the curve, so the sideways push the phone feels, compared with which way the gyroscope says it is turning, shows where “forward” is. Written as complex numbers,

$$\hat{\beta} = \arg \sum_t \frac{a_t}{|a_t|} \overline{i v_t \omega_t} \quad (2)$$

where a_t is the sideways and forward acceleration measured relative to the compass, v_t the speed and ω_t the turn rate. The phone makes a first estimate after 30 seconds of riding and keeps improving it; when it improves, the part of the ride already drawn is redrawn with the better value.

Walking is handled differently: over four seconds, the back-and-forth of the steps lines up with the direction of walking, and each push-off is sharper than each landing, which says which end of that line is forward.

The compass reading itself. The phone’s built-in heading is unreliable in one common pocket position: when the phone sits head-down at an angle, it scattered by about 49° a minute even though the phone was not moving in the pocket. So the system computes its own heading from the phone’s orientation, using whichever edge of the phone is closest to level. That heading scattered by about 1° a minute in the same position.

Only the first point uses GPS. After that, each step is laid down from the previous point, along the current direction, for the distance the speed covers in that second.

2 How it was tested

GPS was recorded alongside every test and never used by the system; it serves only as the answer key. So that every result describes the newest version, the newest method was run again over all earlier recordings, which were made with the phone’s raw sensor readings saved:

- **Speed.** The speed model was rebuilt exactly as the app builds it — the same fingerprint, the same 12-neighbour average, the same refusals, the same correction — with a memory of 4000 examples drawn at random from every *other* journey, so that no journey was ever measured with its own GPS in the memory. On the journeys the app had recorded with this version itself, the rebuilt model’s distance was within about 8% of what the app had drawn.
- **Direction.** The direction method, including the redrawing, was run over the saved motion sensor readings.

That gives 70 journeys for speed and distance (18 hours, motorcycle and car, the phone in a pocket or lying flat), 64 for whole routes, 62 for direction alone, seven walks long and straight enough to check, and one flight.

In a pocket GPS is itself only accurate to 10–40 m, so it was used over stretches long enough for that not to matter:

- **Speed and distance** are compared over half-minute stretches between GPS fixes that are better than 40 m: the straight distance GPS moved, against the distance the system added.
- **Direction** is compared every half-minute: the bearing from the GPS position 15 s earlier to the one 15 s later, against the system’s heading over the same half-minute.
- **Whole routes** are compared by finding the rotation and the size change that best lay the drawn route over the GPS track, pinned at the shared starting point. The rotation is how far the whole route is turned; the size change is how much too big or too small it is.

3 Results

3.1 Speed

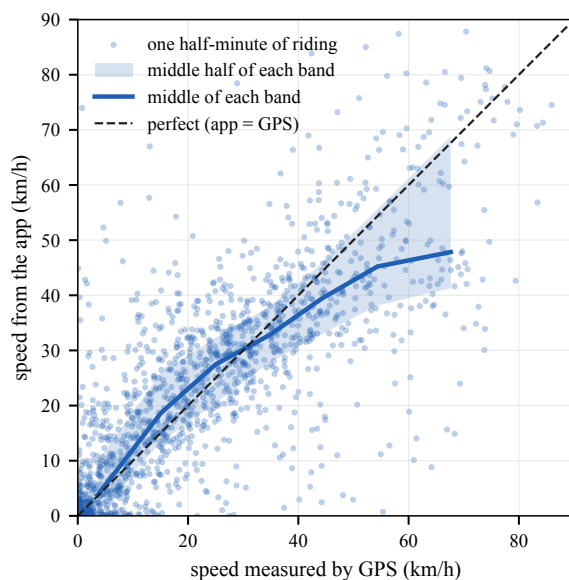


Figure 1: Speed from the system against speed from GPS, over 1457 half-minutes of riding in 70 journeys. On the dashed line the two agree. The solid line is the middle of each 10 km/h band, and the shading holds the middle half of the readings.

Figure 1 shows the speed. Standing still and crawling are recognised correctly. In between, the system squeezes the range: 10–30 km/h reads about 2–4 km/h high, 30–40 km/h is

close, and above 60 km/h it read about 49 where GPS measured 67. The average error is about 9 km/h. The squeeze comes from averaging twelve examples: when a fingerprint could belong to several speeds, the average lands in the middle.

3.2 Distance

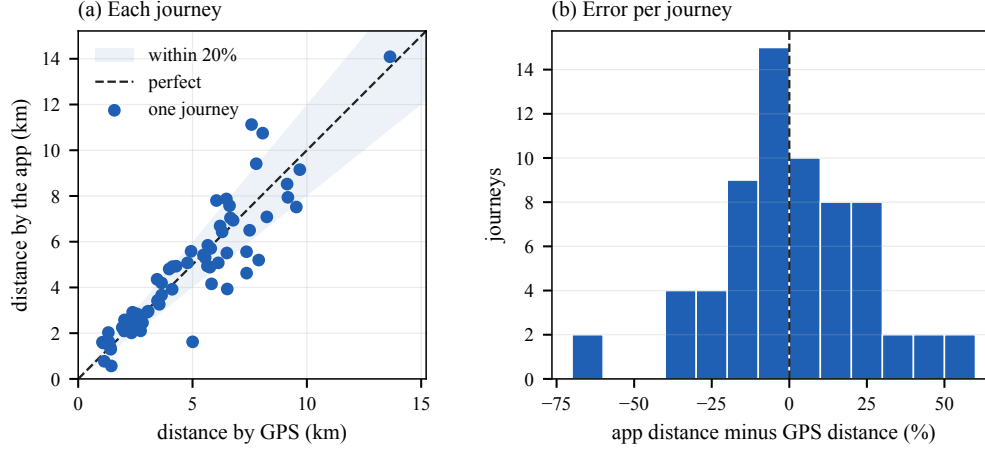


Figure 2: Distance for each of the 66 journeys with at least 1 km that GPS was good enough to check. (a) Each dot is one journey; on the dashed line the app and GPS agree. (b) How far each journey’s distance is from GPS.

The over-reading at low speed and under-reading at high speed largely cancel over a journey. Over the stretches GPS could check — 318 km in all — the system added 314 km, 1.5% short. The middle journey was exactly right; 38% of the journeys were within 10% and 64% within 20% (Figure 2). The worst were 68% short and 54% long.

3.3 Direction

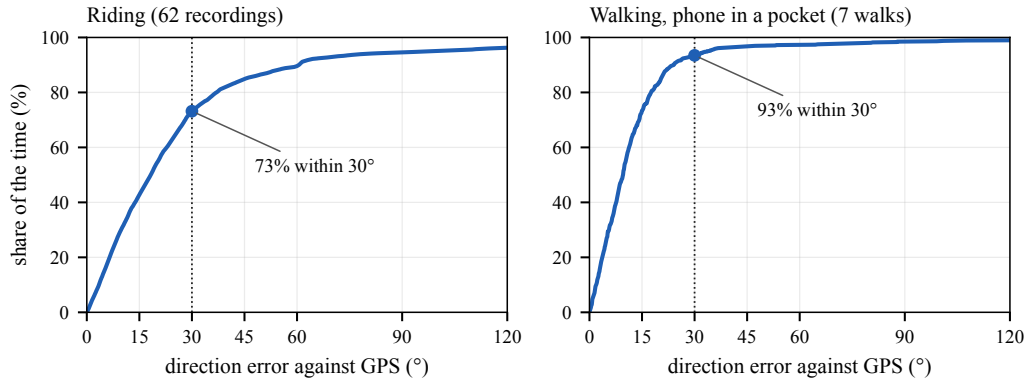


Figure 3: How often the direction is within a given angle of GPS. Left: 20,317 seconds of riding in 62 recordings. Right: 721 seconds of walking in seven walks, phone in a pocket.

While riding, the direction was within 30° of GPS 73% of the time and within 45° 85% of the time; the middle error was 18° (Figure 3). While walking with the phone in a pocket it was within 30° 93% of the time, with a middle error of 10° .

3.4 Whole routes

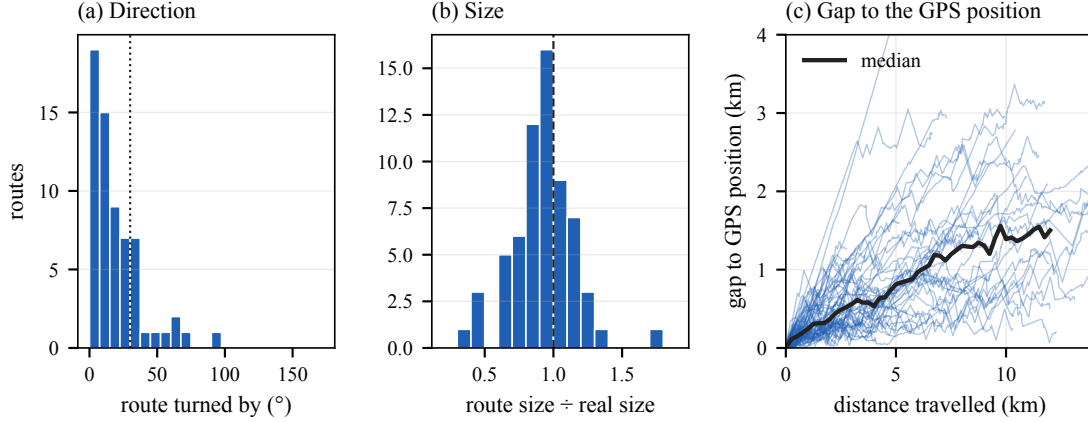


Figure 4: 64 routes drawn by the system, compared with GPS. (a) How far each whole route is turned from the GPS track; the dotted line is 30° . (b) Its size relative to the GPS track. (c) How far the drawn position is from the GPS position as the journey goes on: one thin line per route, and the median.

Table 1: Summary: the newest version against GPS.

	Measured over	Result
Distance	70 journeys, 318 km	1.5% short in total; 64% of journeys within 20%
Speed	1457 half-minutes	average error 9 km/h
Direction, riding	62 recordings	within 30° 73% of the time
Direction, walking	7 walks	within 30° 93% of the time
Direction, flight	1 flight, 15 min	within 30° 99% of the time
Route turned	64 routes, 472 km	median 13° ; 78% within 30°
Route size	64 routes	median 0.95; 69% within 20%
Position gap	64 routes	median 17% of the distance travelled

Figure 4 and Table 1 put speed and direction together. The median route is turned 13° from the truth; 53% are turned less than 15° and 78% less than 30° . Drawn routes come out a little small — a median 0.95 of the real size — even though distance is about right, partly because small direction errors make the drawn path fold slightly, so it ends nearer the start. The drawn position drifts away from GPS as the journey goes on, by a median of about 17% of the distance travelled: about 1.4 km after 10 km.

Six routes are turned more than 45° . Four of them are short journeys, under 3 km, with too few turns to learn the pocket angle well.

3.5 A flight

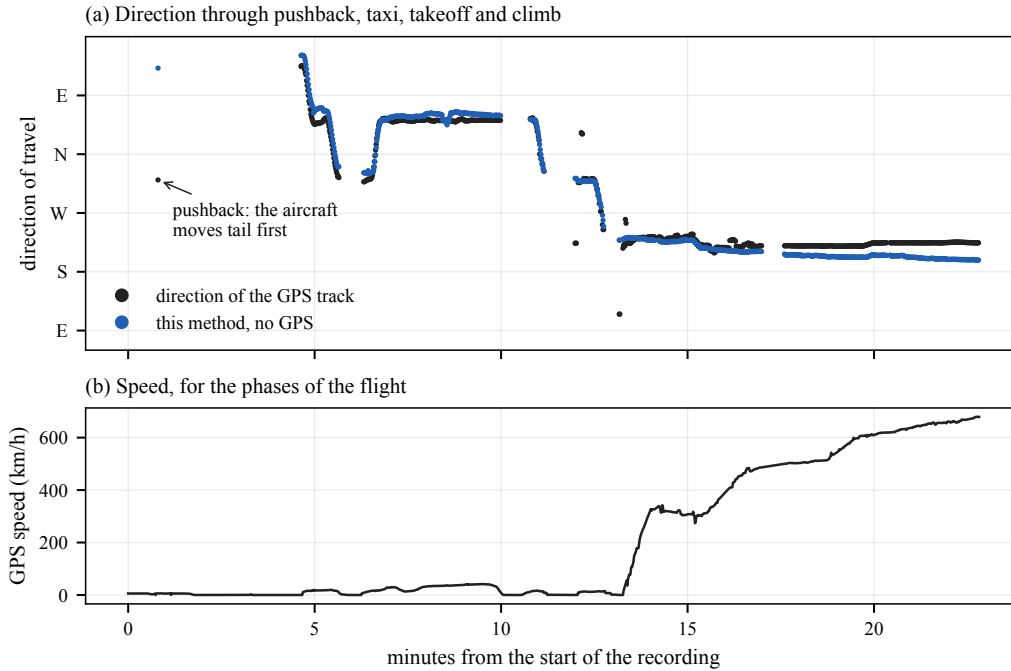


Figure 5: One flight, from the gate to the climb. (a) The direction of travel from the GPS track (black) and from this method, which uses no GPS (blue). (b) The aircraft’s speed from GPS, to show where it was taxiing, taking off and climbing.

Figure 5 follows one flight with the same direction method as on the road. Speed cannot be measured in an aircraft (Section 4), so only direction is compared. While taxiing, the middle error was 8° ; in the air it was 16° ; over the whole recording the direction was within 30° of the GPS track 99% of the time. The one large miss is the pushback at the start, when the aircraft moves tail first: the method gives the way the nose points, which is opposite to the way it moves. In the climb the gap grows slowly, to about 24° by the end of the recording. Part of that may be wind, which pushes an aircraft sideways so that its nose and its track differ; GPS alone cannot tell the two apart.

4 Where it goes wrong

- **Fast riding reads slow, slow riding reads fast.** Above 60 km/h the speed can be half the true value (Figure 1). Over a whole journey the two partly cancel, but a journey that is mostly fast comes out short.
- **The first minute of a ride.** Until the pocket angle is learned, the direction comes from the compass alone, which in a pocket can be far off. The redrawing fixes the saved route afterwards, but a very short ride may never learn the angle well.
- **A phone that moves.** The pocket angle is assumed to stay the same for the whole

ride. If the phone shifts, the rest of the ride is turned by about as much as it moved. When the phone is taken out of the pocket, the hand's movement hides the vehicle's vibration, so the last speed is held; on one ride, 48 km/h was held for ten seconds after the motorcycle had stopped, about 130 m that was never travelled.

- **Car-park ramps.** A car crawling up a concrete ramp shakes like one driving at 40 km/h, so distance is not counted while climbing and turning together. The same test can fire on an ordinary rising, curving road: once for nine seconds at 48 km/h, about 115 m lost.
- **Aircraft.** Speed cannot be measured in an aircraft. The phone's own orientation filter treats the long, steady push of a takeoff as a tilt [4], and the signal is gone. The direction still works (Figure 5).

5 What did not work

Many ideas were built and tested against GPS before the current version. The most instructive:

- **Neural networks.** A network that reads 20–40 seconds of fingerprints at once was more accurate on test data (8.8 against 10.6 km/h) and ran in the app for a while, but after it was retrained it under-read a pocketed ride by a third, and it was removed. Another network, better than the current system on familiar data, was confidently wrong on a flight it had never seen, recording 5% of the distance, where the current system simply declines to answer.
- **Direction from speeding up and braking.** The push when accelerating in a straight line points forward, so it should give direction. It was 44° off; turns work far better.
- **Other direction ideas:** the axis the motorcycle leans about (25° off), the direction of the rider's thigh (73° off), and the swing of the thigh for telling forward from backward when walking (wrong more often than right).
- **How examples were forgotten.** When the store was full, it used to throw away examples from whichever speed it held most of, to keep rare fast examples. That left it holding 5% of its examples below 5 km/h when 24% of real riding is that slow, so slow riding read far too fast. Throwing away a random example instead fixed most of it.

6 Lessons

- **The phone's own definitions caused more error than the models.** Its built-in heading becomes unreliable in a common pocket position, and its “user acceleration” points the opposite way to the vehicle's acceleration. Each of these, once found, mattered more than any change to the estimators.

- **A test must not use what it is testing without.** Twice during development the no-GPS test mode quietly used GPS for direction, which made its results look better than they were.
- **Check what a learning system remembers.** The store’s way of forgetting examples was sensible on its own and still shifted every answer, and it went unnoticed until its contents were compared with the riding they came from.

Limits of this evaluation

All recordings were made by one person, on one iPhone, mostly in one city, by motorcycle, car, on foot and on one flight; other people, phones and vehicles may behave differently. Most results come from running the newest method again over earlier recordings rather than from live use, though on the journeys the app recorded with this version the rerun matched it closely. GPS in a pocket is itself only accurate to tens of metres, so small errors cannot be measured reliably.

References

- [1] A. Cohen and I. Klein. CarSpeedNet: Learning-based speed estimation from accelerometer-only inertial sensing. arXiv:2401.07468, 2024.
- [2] Smartphone MEMS accelerometer and gyroscope measurement errors: laboratory testing and analysis of the effects on positioning performance. *Sensors*, 23(17):7609, 2023.
- [3] Pedestrian dead reckoning with novel heading estimation under magnetic interference and multiple smartphone postures. *Measurement*, 183, 2021.
- [4] Quaternion-based complementary filter for attitude determination of a smartphone. *IEEE Sensors Journal*, 16(15), 2016.